Team Data Science Practicum (DATA 481)

Date: February 19, 2026

To: Zhihui (Duke University), Dr. Jingping Nie (UNC Chapel Hill), and Dr. J. S. Marron (UNC Chapel Hill)

From: The Consulting Team (Arth Vijay, Team Lead)

Subject: Comprehensive Project Scope, Team Charter, and Strategic Roadmap

---

1. Executive Summary

Our team is engaged in a **Team Data Science Practicum** designed to bridge the gap between theoretical data analysis and real-world scientific challenges. As a "consulting team," we are tasked by our primary client, Zhihui, with decoding high-fidelity network signal waveforms into actionable labeled information. This project moves beyond standard homework assignments to deliver a robust, reproducible data pipeline and predictive machine learning models.

---

2. Stakeholder Matrix

Stakeholder	Role	Key Expectations
Zhihui (Duke)	**Primary Client**	Accuracy in decoding LSIG bits into packet properties (transmission time, data rate) using specific waveform inputs.
Dr. Nie & Dr. Marron	**Guiding Instructors**	Oversight of team leadership, task division, and the delivery of results that demonstrate clear scientific and communication insight.
Arth Vijay	**Team Leader**	Facilitating group discussion, managing the bridge between client and team, and monitoring overall progress.

Consulting Team	**Delivery Team**	(Daniel, Pranav, Mihir, Kerry, Muhammad, Teja) Executing ML/DL modeling and ensuring a team result superior to individual effort.

---

3. Project Scope & Machine Learning Mandate

The core objective of this engagement is to translate raw **L-Signal (LSIG)** bits from a **7.1GB dataset** (5 MHz sampling rate) into actionable packet properties. This process is divided into two distinct phases: initial data engineering and high-accuracy predictive modeling.

The Machine Learning Approach

Our strategy moves beyond traditional signal processing by leveraging **Machine Learning (ML)** to identify complex patterns within high-rate waveforms. Because network signals are often subject to noise and environmental interference, standard deterministic decoding may fail. By training models on specific segments of the signal, **packetHigh** and **packetLow**, we can "teach" the system to recognize the underlying signatures of data rates and transmission times. This involves transforming raw signal amplitudes into feature vectors that algorithms like **Support Vector Machines (SVM)** and **Random Forest** can use to classify packet types with high precision.

Technical Pain Points & Constraints

- **Data Volume:** Efficiently managing large `.mat` files within computational environments to process the 7.1GB sub-dataset without memory overflow.
- **Decoding Logic:** Accurately mapping binary tuples to specific N_DBPS values and calculating transmission times using the standard formula:

$$16 + (num_data_symbols \times 16)$$

- **Parity Validation:** Implementing rigorous logic to ensure the parity bit correctly reflects the sum of bits, maintaining data integrity across the pipeline.

Phase 2: Workstream Breakdown & Modeling

This framework governs our transition from data preparation to deploying high-accuracy models.

- **Workstream (i) - Team A (Daniel, Mihir, Teja):** Utilizing **packetHigh** as the primary feature set to predict specific **LSIG bits**.
- **Workstream (ii) - Team B (Pranav, Muhammad, Kerry):** Utilizing **packetLow** as the feature set to predict **Packet Properties** such as Data Rate and Transmission Time.

Tech Stack & Core Deliverables

To ensure professional-grade delivery, the team utilizes a standardized suite of tools and produces executive-ready outputs:

- **Tools:** Python and R for data analysis and modeling; GitHub for version control and collaborative coding.
- **Model Architectures:** Implementation of **SVM**, **Random Forest**, and **Deep Learning** techniques to compare performance across the waveform segments.
- **The "Killer Graph":** A single, narrative-driven visual result designed for executive communication that summarizes our best predictive outcome with minimal cognitive load.
- **Final Team Report:** A comprehensive document detailing the methodology, ML results, and scientific insights for the stakeholders.

---

4. Strategic Roadmap & Milestones

Phase 1: Foundation & Pilot Testing

- **Interval 1 (Jan 22 – Feb 5): Foundation & Repository Setup**
  - **Version Control:** Established the central GitHub repository for collaborative development.
  - **Decoding Logic:** Built the initial decoding script to map `rate_bits` to protocol maps (BPSK, 16-QAM) for signal interpretation.
- **Interval 2 (Feb 5 – Feb 19): Protocol Filtering & Pilot Testing**
  - **Data Refinement:** Focused on Non-HT packet filtering and finalized the `decoded_results_nonht_only.csv` dataset.
  - **Pilot ML Models:** Commenced initial machine learning testing using Logistic Regression and SVM on a controlled subset of 1,000 samples to establish a performance baseline.
- **Interval 3 (Feb 19 – Mar 5): Metric Optimization & Decoding Refinement**
  - **Performance Tuning:** Refined ML practices specifically on the data subset, focusing on feature engineering and hyperparameter tuning.
  - **Validation:** Achieved significantly improved accuracy metrics and test results on the subset, confirming the viability of the ML-driven decoding approach.

Phase 2: Infrastructure & Scaling

- **Interval 4 (Mar 5 – Mar 12): Infrastructure Planning & Preparation**
  - **Scalability Assessment:** Conducted technical requirements analysis for processing the full dataset (64GB+), identifying the need for cloud-based compute resources.

    - **Documentation:** Finalized internal documentation for the decoding logic to ensure consistency during the upcoming infrastructure migration.
- **[Spring Break: March 13 – March 23]**
    - *Project Recess: No active development.*
- **Interval 5 (Mar 24 – Apr 7): Azure Migration & Cloud Infrastructure**
    - **Azure Migration:** Successfully migrated the project architecture and data assets to the Microsoft Azure cloud environment.
    - **VM Creation & Hosting:** Configured and deployed a high-performance Azure Virtual Machine (VM) to serve as the centralized environment for full-scale data processing.
    - **Full Dataset Decoding:** Executed the refined decoding logic across the entire dataset within the VM, successfully labeling all available packets for the final pipeline.

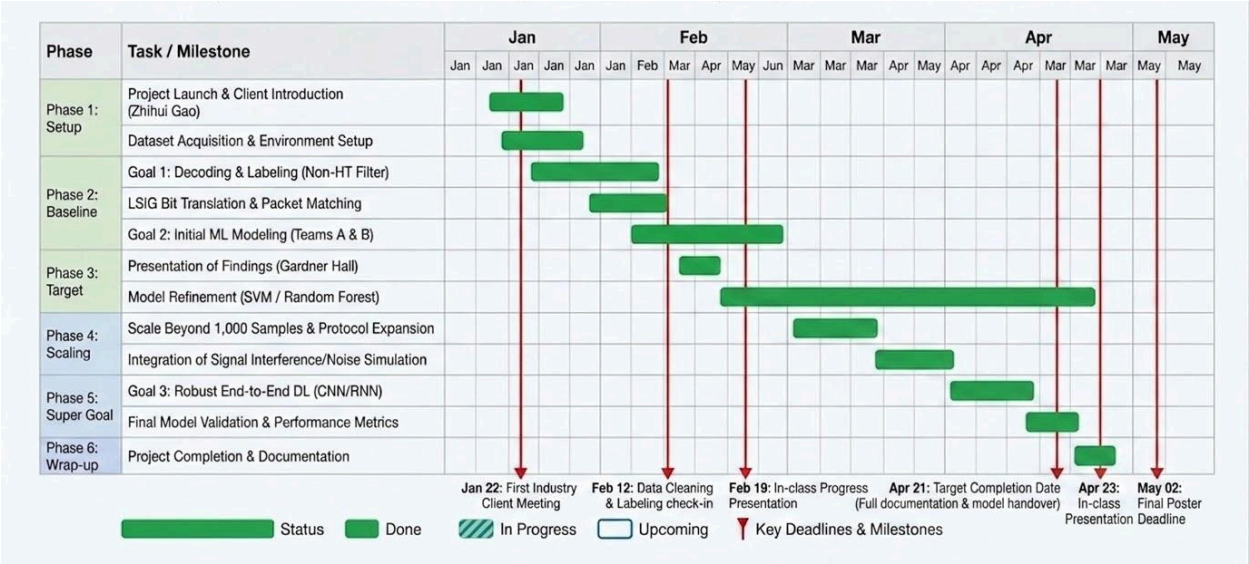
Phase 3: Integration & Final Delivery

- **Interval 6 (Apr 7 – Apr 14): ML Pipeline Integration & Advanced Modeling**
    - **Cloud Pipeline Deployment:** Established automated ML pipelines within the Azure environment to streamline the flow from raw waveform to labeled output.
    - **Subgroup Model Training:** Initiated parallel training sessions for **Team A** (High Waveform to LSIG) and **Team B** (Low Waveform to Properties) using the complete, cloud-hosted dataset.
- **Interval 7 (Apr 14 – Apr 21): Final Integration & Result Consolidation**
    - **End-to-End Completion:** Finalized the comprehensive "unzipping-through-analysis" pipeline, ensuring seamless data flow and predictive consistency.
    - **Metric Finalization:** Consolidated final performance metrics (Accuracy, F1-scores) for both high and low-rate models for the final stakeholder delivery.
    - **Offboarding & Documentation:** Completed VM configuration documentation and finalized the slide structure for the project close-out.
- **Interval 8 (Apr 23 – May 2): Final Consolidation & Professional Delivery**
    - **Poster Design & Production:** Finalized the visual representation of the project, including architecture diagrams of the Azure pipeline and ML performance charts.
    - **Technical Report & Slide Finalization:** Conducted a final audit of the predictive models to ensure the presentation for the industry client (Zhihui Gao) was technically rigorous and grounded in the finalized metrics.

Deliverables Schedule

Date	Milestone	Requirement
Mar 12	**Team Status Update**	Present initial ML scaling results in Gardner Hall; mid-semester review of accuracy.
Apr 23	**Final Team Update**	Presentation of finalized model comparisons and cross-functional results to the full class.
May 2	**Final Presentation**	Professional poster session and delivery of the final technical report at SDSS Manning (4:00 – 7:00 PM).

Gantt Chart:



---

5. Norms and Success Metrics

Behavioral Norms & "Virtuous Cycle"

- **GitHub Workflow:** All members must follow the standard procedure: **Pull** → **Work** → **Commit with Comment** → **Push**.

- **Circle of Effort:** We utilize a task-based rotation (Coder → Reviewer → Supporter) to ensure cross-team technical understanding.
- **Meeting Protocol:** Bi-weekly check-ins with the client are mandatory. For missed classes due to illness or interviews, members must join via Zoom if possible and notify the team and professors in advance.
- **AI Policy:** Large Language Models (e.g., Gemini, ChatGPT) may be used for productivity. However, members are responsible for any "hallucinations" and must include a dedicated section in write-ups describing their use.

Success Metrics

Performance is evaluated on collective clarity and team interaction rather than individual "smartest person" status.

- **Team Synergy:** Is the team result demonstrably better than what individuals could have achieved alone?
- **Engagement:** Consistency in attendance, participation in discussions, and timely response to peer evaluation surveys.
- **Clarity:** The effectiveness and professional quality of the final report and oral presentations.